

Clinical Risk Assessment Report: Patient 58

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 32.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.47x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +6.9%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.38 [Bottom 6%] (+6.9%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.36 [Avg (66th %ile)] (+4.0%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.42 [Bottom 10%] (+3.2%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.79 [Top 25%] (+2.5%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.22 [Bottom 11%] (+1.8%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

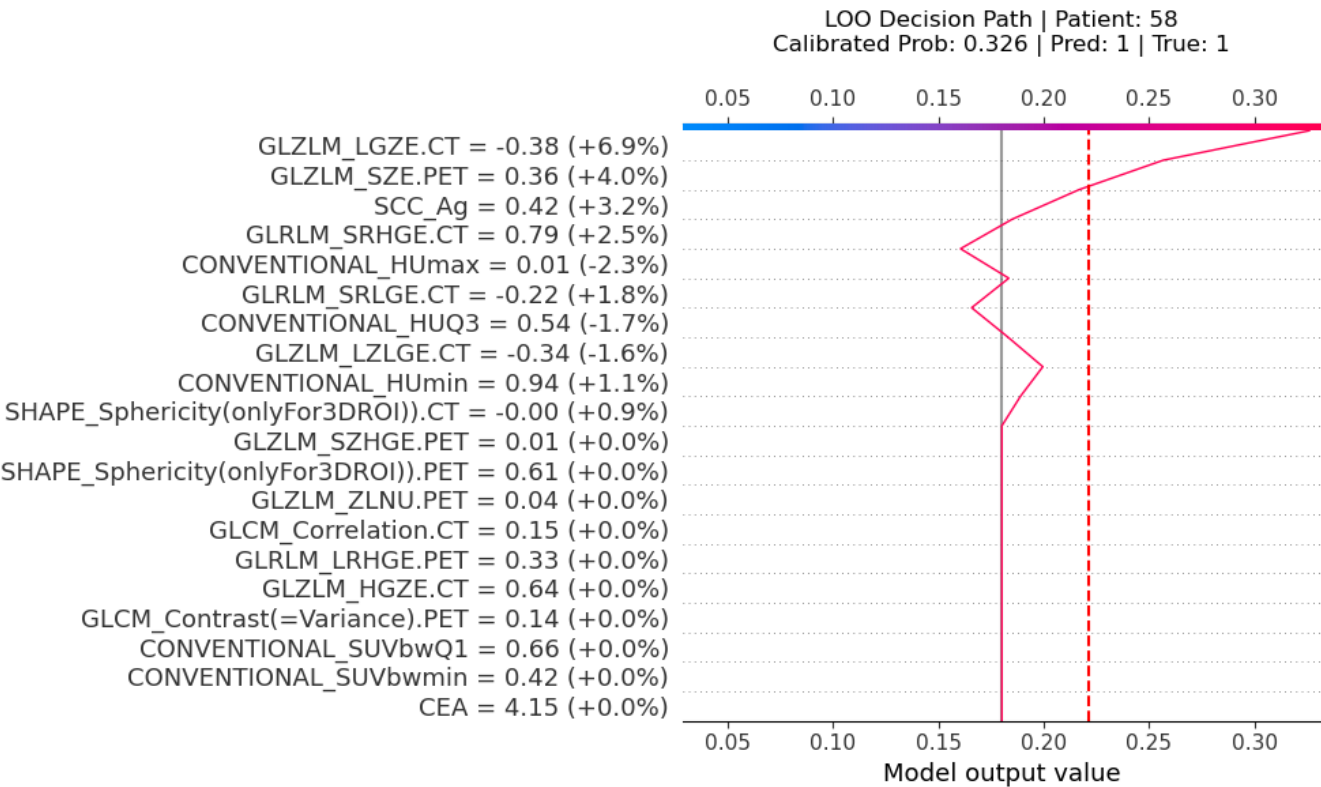
3. Protective / Mitigating Factors (Reducing Risk)

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.01 [Avg (46th %ile)] (-2.3%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.54 [Avg (61th %ile)] (-1.7%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.34 [Bottom <1%] (-1.6%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

